

ASSIGNMENT:-08

Name: Rajdeep Das

Roll No: CSB25203

Subjec: Advanced Programming

```
import { useState, useMemo, useEffect } from "react";
```

```
let nextId = 4;
```

```
const initialMap = new Map([
```

```
  [1, { id: 1, name: "Aria Chen", enrolledCourses: new Set(["Algorithms", "Machine Learning",  
"Linear Algebra"])}, gpa: 3.9 ]],
```

```
  [2, { id: 2, name: "Marcus Webb", enrolledCourses: new Set(["Algorithms", "Systems  
Programming", "Networks"])}, gpa: 3.5 ]],
```

```
  [3, { id: 3, name: "Priya Nair", enrolledCourses: new Set(["Machine Learning", "Statistics",  
"Linear Algebra"])}, gpa: 3.7 ]],
```

```
]);
```

```
const FIRST_NAMES =
```

```
["Aria","Marcus","Priya","James","Sofia","Liam","Emma","Noah","Olivia","Ethan","Ava","Mason","Isabe  
lla","Logan","Mia","Lucas","Charlotte","Aiden","Amelia","Jackson","Harper","Sebastian","Evelyn","Mat  
eo","Abigail","Jack","Emily","Owen","Elizabeth","Theodore","Mila","Asher","Ella","Henry","Riley","Alexa  
nder","Zoey","Daniel","Nora","Michael","Lily"];
```

```
const LAST_NAMES =
```

```
["Chen","Webb","Nair","Smith","Johnson","Williams","Brown","Jones","Garcia","Miller","Davis","Wilso  
n","Taylor","Anderson","Thomas","Jackson","White","Harris","Martin","Thompson","Young","Lee","Wa  
lker","Hall","Allen","King","Scott","Green","Baker","Adams"];
```

```
// — Complexity Panel —————
```

```
function ComplexityPanel({ allCourses }) {
```

```
const sizes=[100,200,400, 600, 800, 1000];
```

```
const rows=useMemo(() => {  
  const courses=[...allCourses];  
  if(!courses.length)return [];
```

```
  return sizes.map(n => {  
    //Generate random students  
    const students=Array.from({ length: n }, () => {  
      const num=1+Math.floor(Math.random() * Math.min(4, courses.length));  
      const shuffled=[...courses].sort(() => Math.random() - 0.5);  
      return{name: `S${n}`, gpa: Math.random() * 2 + 2, enrolledCourses: new  
Set(shuffled.slice(0,num)) };  
    });  
  
    //Measure time to filter by a random course (repeat 50x for stable ms reading)  
    const course=courses[Math.floor(Math.random() * courses.length)];  
    const REPS = 50;  
    const t0=performance.now();  
    for(let r=0;r<REPS;r++) {  
      students.filter(s=>s.enrolledCourses.has(course));  
    }  
    const ms=(performance.now() - t0) / REPS;  
  
    return{n,ms:ms<0.01 ? "<0.01" : ms.toFixed(3) };  
  });  
}, [allCourses]);
```

```
const thStyle = {  
  fontSize:10,color:"#aaa", fontWeight: 500,  
  padding:"6px8px",borderBottom: "1px solid #f0f0f0", whiteSpace: "nowrap",
```

```

};

consttdStyle = {
  fontSize:12, color: "#555",
  padding:"6px 8px", borderBottom: "1px solid #f5f5f5",
};

return (
  <divstyle={{ background: "#fff", borderRadius: 10, padding: 18, border: "1px solid #e8e8e8" }}>
    <divstyle={{ display: "flex", justifyContent: "space-between", alignItems: "baseline",
marginBottom: 8 }}>
      <pstyle={{ fontSize: 12, fontWeight: 600, color: "#555", textTransform: "uppercase",
letterSpacing: "0.06em", margin: 0 }}>
        TimeComplexity
      </p>
      <spanstyle={{ fontSize: 12, fontWeight: 700, fontFamily: "monospace", color: "#3b82f6"
}}>O(n)</span>
    </div>
    <pstyle={{ fontSize: 11, color: "#aaa", marginBottom: 12, lineHeight: 1.5 }}>
      Filterstudents by course · avg over 50 runs
    </p>

    <tablestyle={{ width: "100%", borderCollapse: "collapse" }}>
      <thead>
        <tr>
          <thstyle={{ ...thStyle, textAlign: "left" }}>Students (n)</th>
          <thstyle={{ ...thStyle, textAlign: "right" }}>Time (ms)</th>
        </tr>
      </thead>
      <tbody>
        {rows.map(row => (
          <trkey={row.n}>
            <tdstyle={{ ...tdStyle, textAlign: "left", color: "#111", fontWeight: 500 }}>{row.n}</td>

```

```

        <td style={{ ...tdStyle, textAlign: "right", fontFamily: "monospace", color: "#111"
}}>{row.ms}</td>

      </tr>

    )}

  </tbody>

</table>

</div>

);
}

```

//—Main App —————

```

export default function App() {
  const [studentsMap, setStudentsMap] = useState(initialMap);
  const [filterCourse, setFilterCourse] = useState("");
  const [form, setForm] = useState({ name: "", courses: "", gpa: "" });
  const [error, setError] = useState("");

  useEffect(() => {
    document.documentElement.style.cssText =
    "width:100%;height:100%;margin:0;padding:0;background:#f4f4f4;";

    document.body.style.cssText = "width:100%;min-
height:100vh;margin:0;padding:0;background:#f4f4f4;";

    const root = document.getElementById("root");
    if(root) root.style.cssText = "width:100%;min-height:100vh;background:#f4f4f4;";
  }, []);

  const allStudents = useMemo(() => [...studentsMap.values()], [studentsMap]);

  const allCourses = useMemo(() =>
    allStudents.reduce((acc, s) => { s.enrolledCourses.forEach(c => acc.add(c)); return acc; },
    new Set()),

```

```
[allStudents]
```

```
);
```

```
const displayed = useMemo(() => {  
  const filtered = filterCourse  
    ?allStudents.filter(s => s.enrolledCourses.has(filterCourse))  
    :allStudents;  
  return [...filtered].sort((a, b) => b.gpa - a.gpa);  
},[allStudents, filterCourse]);
```

```
function addStudent() {  
  const name = form.name.trim();  
  const gpa = parseFloat(form.gpa);  
  const courses = form.courses.split(",").map(c => c.trim()).filter(Boolean);  
  if(!name) return setError("Name required.");  
  if(isNaN(gpa)) return setError("Enter a valid GPA.");  
  if(!courses.length) return setError("At least one course required.");  
  setError("");  
  const id = nextId++;  
  setStudentsMap(prev => new Map([...prev, [id, { id, name, enrolledCourses: new Set(courses),  
gpa}]]));  
  setForm({ name: "", courses: "", gpa: "" });  
}
```

```
function removeStudent(id) {  
  setStudentsMap(prev => { const next = new Map([...prev]); next.delete(id); return next; });  
}
```

```
const inputStyle = {  
  width: "100%", padding: "8px 10px", fontSize: 13,  
  background: "#f9f9f9", border: "1px solid #e0e0e0",
```

```
borderRadius: 6, outline: "none", color: "#111",
marginBottom: 8, boxSizing: "border-box",
};
```

```
return (
```

```
<divstyle={{
```

```
  fontFamily: "'Inter', system-ui, sans-serif",
```

```
  background: "#f4f4f4", minHeight: "100vh", width: "100vw",
```

```
  boxSizing: "border-box", padding: 32,
```

```
  position: "absolute", top: 0, left: 0,
```

```
}}>
```

```
<style>`
```

```
  @import
```

```
url('https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600&display=swap');
```

```
  *,*::before,*::after { box-sizing: border-box; margin: 0; padding: 0; }
```

```
  html,body,#root { width: 100% !important; min-height: 100vh !important; background:
  #f4f4f4 !important; }
```

```
`</style>
```

```
<divstyle={{ marginBottom: 28 }}>
```

```
  <h1style={{ fontSize: 20, fontWeight: 600, color: "#111", margin: 0 }}>Course
  Enrollment</h1>
```

```
  <pstyle={{ fontSize: 13, color: "#888", marginTop: 4 }}>
```

```
    {allStudents.length} students · {allCourses.size} unique courses
```

```
  </p>
```

```
</div>
```

```
<divstyle={{ display: "grid", gridTemplateColumns: "280px 1fr", gap: 24, alignItems: "start" }}>
```

```
  <divstyle={{ display: "flex", flexDirection: "column", gap: 16 }}>
```

```
    {/*Add student */}
```

```

<div style={{ background: "#fff", borderRadius: 10, padding: 18, border: "1px solid #e8e8e8"
}}>

  <p style={{ fontSize: 12, fontWeight: 600, color: "#555", textTransform: "uppercase",
letterSpacing: "0.06em", margin: "0 0 12px" }}>Add Student</p>

  <input style={inputStyle} placeholder="Name" value={form.name} onChange={e =>
setForm(p => ({ ...p, name: e.target.value })))} />

  <input style={inputStyle} placeholder="GPA (any number)" type="number"
value={form.gpa} onChange={e => setForm(p => ({ ...p, gpa: e.target.value })))} />

  <input style={inputStyle} placeholder="Courses, comma separated" value={form.courses}
onChange={e => setForm(p => ({ ...p, courses: e.target.value })))} />

  {error && <p style={{ fontSize: 12, color: "#e55", marginBottom: 8 }}>{error}</p>}

  <button onClick={addStudent} style={{ width: "100%", background: "#111", color: "#fff",
border:"none", borderRadius: 6, padding: "9px 0", fontSize: 13, fontWeight: 500, cursor:
"pointer"}}>

    Add Student

  </button>

</div>

```

```

{/*Filter */}

<div style={{ background: "#fff", borderRadius: 10, padding: 18, border: "1px solid #e8e8e8"
}}>

  <p style={{ fontSize: 12, fontWeight: 600, color: "#555", textTransform: "uppercase",
letterSpacing: "0.06em", margin: "0 0 10px" }}>Filter by Course</p>

  <select value={filterCourse} onChange={e => setFilterCourse(e.target.value)} style={{
...inputStyle, marginBottom: 0, cursor: "pointer" }}>

    <option value="">All courses</option>

    {[...allCourses].sort().map(c => <option key={c} value={c}>{c}</option>)}

  </select>

  {filterCourse && (

    <button onClick={() => setFilterCourse("")} style={{ marginTop: 8, fontSize: 12, color:
"#999",background: "none", border: "none", cursor: "pointer", padding: 0 }}>

      Clear filter ×

    </button>

  )}

</div>

```

```

    { /*Allcourses */}

    <divstyle={{ background: "#fff", borderRadius: 10, padding: 18, border: "1px solid #e8e8e8"
}}>

        <pstyle={{ fontSize: 12, fontWeight: 600, color: "#555", textTransform: "uppercase",
letterSpacing: "0.06em", margin: "0 0 10px" }}>All Courses</p>

        <divstyle={{ display: "flex", flexWrap: "wrap", gap: 6 }}>

            {[...allCourses].sort().map(c => (

                <span key={c} style={{ fontSize: 12, background: "#f0f0f0", color: "#444", padding: "3px
8px",borderRadius: 4 }}>{c}</span>

            ))}

        </div>

    </div>

    <ComplexityPanel allCourses={allCourses} />

</div>

```

```

    { /*Student rows */}

    <div>

        {filterCourse && (

            <pstyle={{ fontSize: 13, color: "#555", margin: "0 0 10px" }}>

                {displayed.length} student{displayed.length !== 1 ? "s" : ""} in

            <strong>{filterCourse}</strong>

            </p>

        )}

        {displayed.length === 0 && (

            <divstyle={{ background: "#fff", borderRadius: 10, padding: 40, textAlign: "center", color:
"#bbb",border: "1px solid #e8e8e8", fontSize: 14 }}>

                Nostudents found.

            </div>

        )}

```



```
>  
  Remove  
</button>  
</div>  
  )}  
</div>  
</div>  
</div>  
</div>  
);  
}
```